

JCB SAVANNAH PLANT - AIR QUALITY UPGRADES

2000 BAMFORD BLVD, POOLER, GA. 31322

LOCATION OF WORK



SHEET INDEX			
SHEET NUMBER	SHEET NAME	SHEET ISSUE DATE	CURRENT REVISION
C000	COVER SHEET	12/1/2025	PRELIMINARY REVIEW SET
A111	NEW PARTITION PLAN	12/1/2025	PRELIMINARY REVIEW SET
A121-ALT	ARCHITECTURAL ROOF PLAN - ALTERNATE 2	12/1/2025	PRELIMINARY REVIEW SET
M000	MECHANICAL SYMBOLS & ABBREVIATIONS	12/1/2025	PRELIMINARY REVIEW SET
M001	MECHANICAL GENERAL SHEET	12/1/2025	PRELIMINARY REVIEW SET
MD111	DEMOLITION MECHANICAL FLOOR PLAN	12/1/2025	PRELIMINARY REVIEW SET
MD121	DEMOLITION MECHANICAL ROOF PLAN	12/1/2025	PRELIMINARY REVIEW SET
M101	TEMPORARY MECHANICAL OVERALL PLAN	12/1/2025	PRELIMINARY REVIEW SET
M111	MECHANICAL OVERALL PLAN	12/1/2025	PRELIMINARY REVIEW SET
M111A	MECHANICAL FLOOR PLAN - AREA A	12/1/2025	PRELIMINARY REVIEW SET
M111B	MECHANICAL FLOOR PLAN - AREA B	12/1/2025	PRELIMINARY REVIEW SET
M111C	MECHANICAL FLOOR PLAN - AREA C	12/1/2025	PRELIMINARY REVIEW SET
M111D	MECHANICAL FLOOR PLAN - AREA D	12/1/2025	PRELIMINARY REVIEW SET
M121	MECHANICAL ROOF PLAN	12/1/2025	PRELIMINARY REVIEW SET
M201	MECHANICAL ENLARGED PLAN - BASE BID	12/1/2025	PRELIMINARY REVIEW SET
M111-ALT	MECHANICAL OVERALL PLAN - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M111A-ALT	MECHANICAL FLOOR PLAN - AREA A - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M111B-ALT	MECHANICAL FLOOR PLAN - AREA B - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M111C-ALT	MECHANICAL FLOOR PLAN - AREA C - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M111D-ALT	MECHANICAL FLOOR PLAN - AREA D - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M121-ALT	MECHANICAL ROOF PLAN - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M201-ALT	MECHANICAL ENLARGED PLAN - ALTERNATE 1	12/1/2025	PRELIMINARY REVIEW SET
M301	MECHANICAL SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET
M401	MECHANICAL CONTROLS	12/1/2025	PRELIMINARY REVIEW SET
M501	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M502	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M503	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M504	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M505	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M506	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M507	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M508	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
M509	MECHANICAL DETAILS	12/1/2025	PRELIMINARY REVIEW SET
P000	PLUMBING SYMBOLS & ABBREVIATIONS	12/1/2025	PRELIMINARY REVIEW SET
P101	PLUMBING UNDERFLOOR PLAN	12/1/2025	PRELIMINARY REVIEW SET
P111	PLUMBING FLOOR PLAN - OVERALL	12/1/2025	PRELIMINARY REVIEW SET
P111A	PLUMBING FLOOR PLAN - AREA A	12/1/2025	PRELIMINARY REVIEW SET
P111B	PLUMBING FLOOR PLAN - AREA B	12/1/2025	PRELIMINARY REVIEW SET
P111C	PLUMBING FLOOR PLAN - AREA C	12/1/2025	PRELIMINARY REVIEW SET
P111D	PLUMBING FLOOR PLAN - AREA D	12/1/2025	PRELIMINARY REVIEW SET
P201	PLUMBING ENLARGED PLANS	12/1/2025	PRELIMINARY REVIEW SET
P301	PLUMBING SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET
P401	PLUMBING RISERS	12/1/2025	PRELIMINARY REVIEW SET
P501	PLUMBING DETAILS	12/1/2025	PRELIMINARY REVIEW SET
E000	ELECTRICAL SYMBOLS & ABBREVIATIONS	12/1/2025	PRELIMINARY REVIEW SET
E500	ELECTRICAL PANEL SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET
E501	ELECTRICAL PANEL SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET
E502	ELECTRICAL PANEL SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET
E503	ELECTRICAL PANEL SCHEDULES	12/1/2025	PRELIMINARY REVIEW SET



PRELIMINARY
REVIEW SET
NOT FOR REGULATORY
APPROVAL, PERMITTING, OR
CONSTRUCTION
WADE A. CLEARY
P.E. REG. NO. 027730
DEC 2, 2025

JCB SAVANNAH PLANT - AIR QUALITY UPGRADES

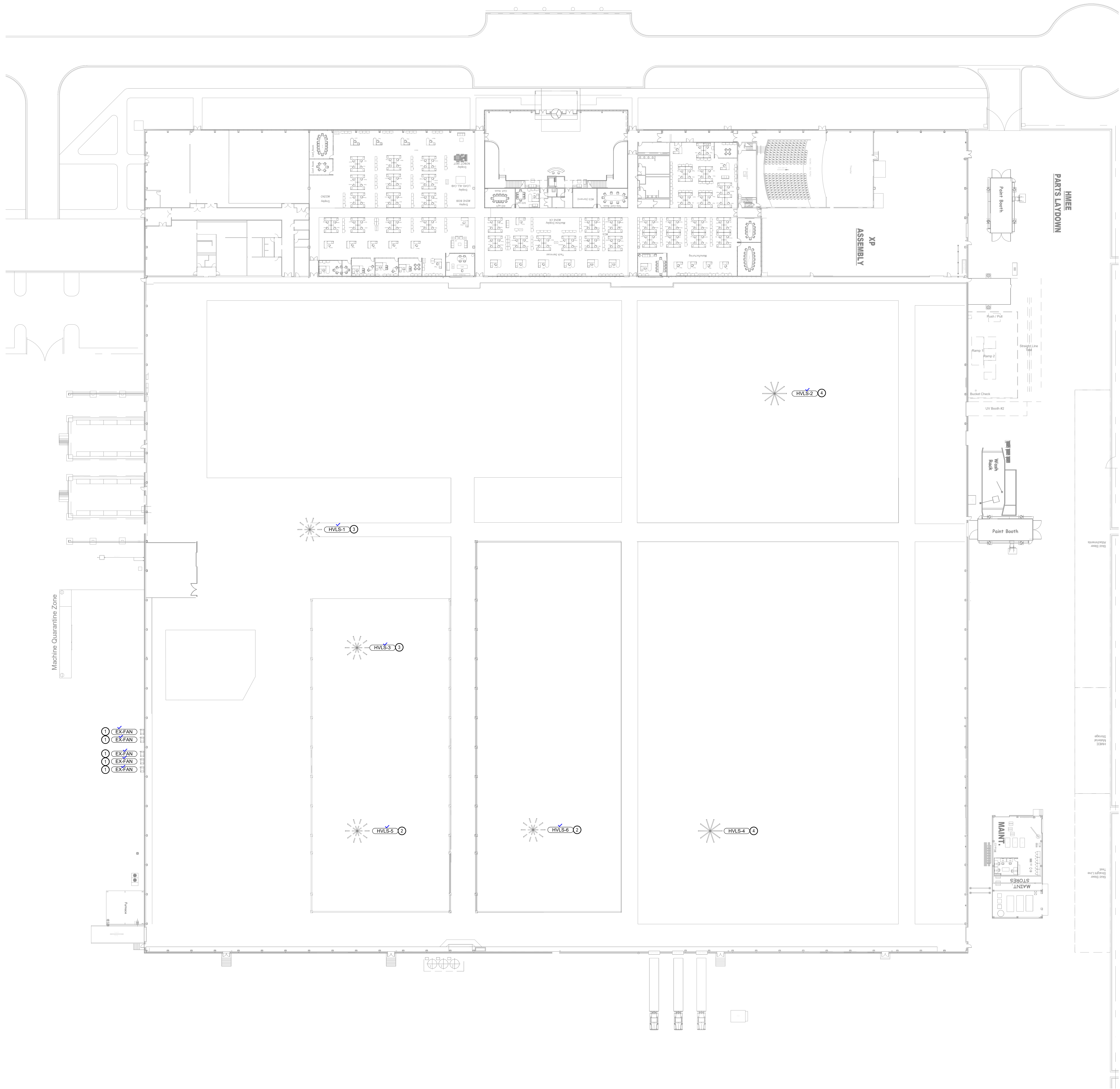
2000 BAMFORD BLVD, POOLER, GA 31322

Project Number	250829(cz)
Date	12/02/2025
Drawn By	Author
Checked By	Checker

Revisions		
No.	Description	Date

DEMOLITION MECHANICAL FLOOR PLAN

MD111



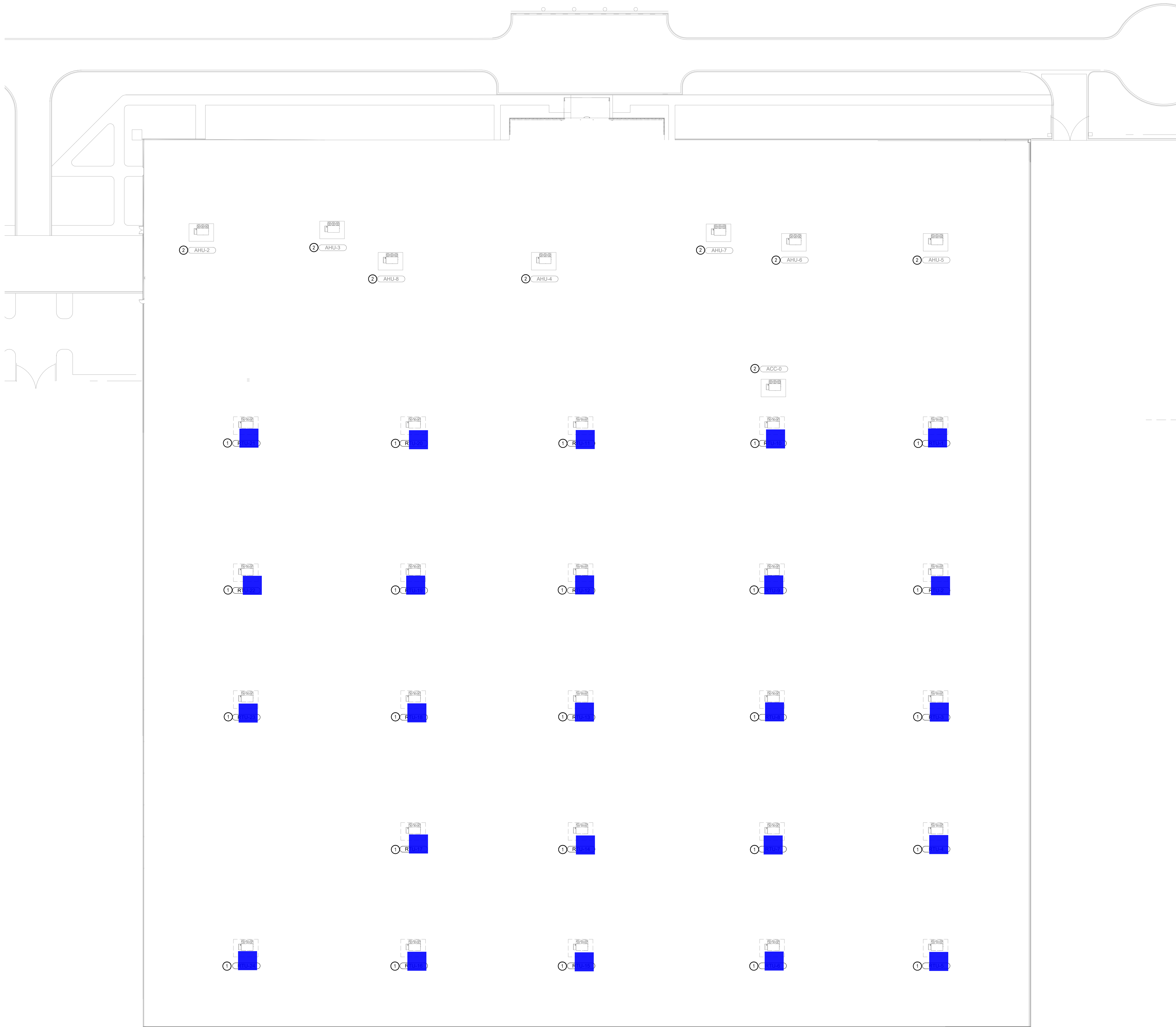
MECHANICAL DEMOLITION KEYED NOTES:		CHECK SM SCOPE
①	DEMOLISH EXISTING FAN, INCLUDING ASSOCIATED ALL CONTROLS, SUPPORTS AND APPURTENANCES. PERMANENTLY GAP AND SEAL PENETRATION INTO BUILDING.	
②	DEMOLISH AND REMOVE EXISTING HVLS FAN, INCLUDING ALL ASSOCIATE CONTROLS, SUPPORTS AND APPURTENANCES.	
③	REMOVE AND SALVAGE EXISTING HVLS FAN. INCLUDE ALL ASSOCIATE CONTROLS, SUPPORTS AND APPURTENANCES.	
④	EXISTING HVLS FAN TO REMAIN.	

0' 15' 30' 60' 120'

SCALE: 1" = 30'-0"

1 DEMOLITION MECHANICAL FLOOR PLAN
MD111 1" = 30'-0"

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- MECHANICAL KEYED NOTES:**
- ① DEMOLISH EXISTING AIR HANDLER, INCLUDING ALL ASSOCIATED CONTROLS, PIPING, AND APPURTENANCES. PERMANENTLY CAP ROOF CURB.
 - ② EXISTING EQUIPMENT TO REMAIN.



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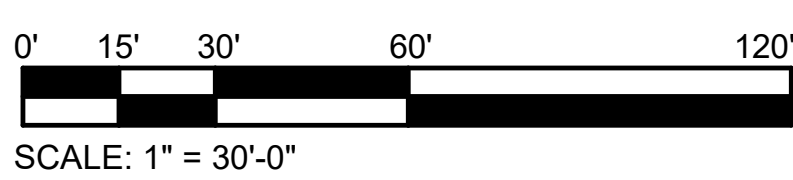
Project Number: 250829(cz)
Date: 12/02/2025
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DEMOLITION
MECHANICAL
ROOF PLAN

MD121

1 DEMOLITION MECHANICAL ROOF PLAN
MD121 1" = 30'-0"



INSULATION SCHEDULE						
System	Location	Material Type	Jacket	Pipe Size	Thickness / Density	Installed Thermal Conductivity k - (R-Value) k - Btu/in(h*1h*2°F)
Return Air Duct	Concealed	Duct Wrap	FSK	N/A	2" / 1#	(R-6)
Return Air Duct	Mechanical Room	Duct Board	FSK	N/A	2" / 3#	(R-6)
Return Air Duct	Outdoors	Duct Board	Aluminum	N/A	3" / 7.5#	(R-6)
Supply Air Duct	Concealed	Duct Wrap	FSK	N/A	2" / 1#	(R-6)
Supply Air Duct	Mechanical Room	Duct Board	FSK	N/A	2" / 3#	(R-6)
Supply Air Duct	Outdoors	Duct Board	Aluminum	N/A	3" / 7.5#	(R-6)
Chilled Water - Exposed	10 ft and below	2.5 # Phenolic	ASJ	All	1.50" / 2.5#	0.21 - 0.27
Chilled Water - All remaining	All remaining locations	Fiberglass Pipe Cover	ASJ	Up to 1"	2.5"	0.21 - 0.27
Chilled Water - All remaining	All remaining locations	Fiberglass Pipe Cover	ASJ	1-1/4" and above	3"	0.21 - 0.27
Condensate	Concealed	Elastomeric	N/A	All	1.00"	0.21 - 0.27
Condensate	Exposed	Elastomeric	Aluminum	All	1.00"	0.21 - 0.27
Condensate	Mechanical Room	Elastomeric	Aluminum	All	1.00"	0.21 - 0.27
Domestic Cold Water	Exterior Walls	Fiberglass Pipe Cover	ASJ	All	1.00"	0.21 - 0.27
Domestic Cold Water	Outdoors	Fiberglass Pipe Cover	Aluminum	All	1.00"	0.21 - 0.27
Domestic Hot Water	Concealed	Fiberglass Pipe Cover	ASJ	0.00" - 1.25" 1.50" - Above	1.00" 1.50"	0.21 - 0.28

PIPING MATERIALS SCHEDULE			
System Name	Sizes	Material Type	Joint Type
Domestic Water - A/G (Plant)	>2"	Type L Copper	Press
Domestic Water - A/G (Plant)	<2"	PEX - type a	Expansion
Natural Gas	<2.5"	Black Iron	Socket Weld
Natural Gas	2.5" & up	Black Iron	Butt Weld
Chilled Water (CHWS/R)	<2"	Type L Copper	Press
Chilled Water (CHWS/R)	2.5" & Up	Sch 40 Carbon Steel	Weld
Compressed Air (CDA)	ALL	Aluminum Blue Pipe	AIR Pipe Conn
Condensate Drains	ALL	Type L Copper	Press
Sweeper Piping	ALL	Sch 40 Carbon Steel	Weld

DUCT CONSTRUCTION SPECIFICATIONS				
Duct System	Pressure Class	Material	Liner - Rect	Liner - Round
Supply Constant Volume	2" wg	Galvanized	None	None
Exposed Duct	2" wg	Galvanized	1"-1.5#	1"-1.5#
Return	2" wg	Galvanized	None	None
Outside Air	2" wg	Galvanized	None	None
Transfer	2" wg	Galvanized	1"-1.5#	None
General Exhaust Plenums	2" wg	Galvanized	None	None
Fume Duct	6" Gavl	Galvanized	None	None



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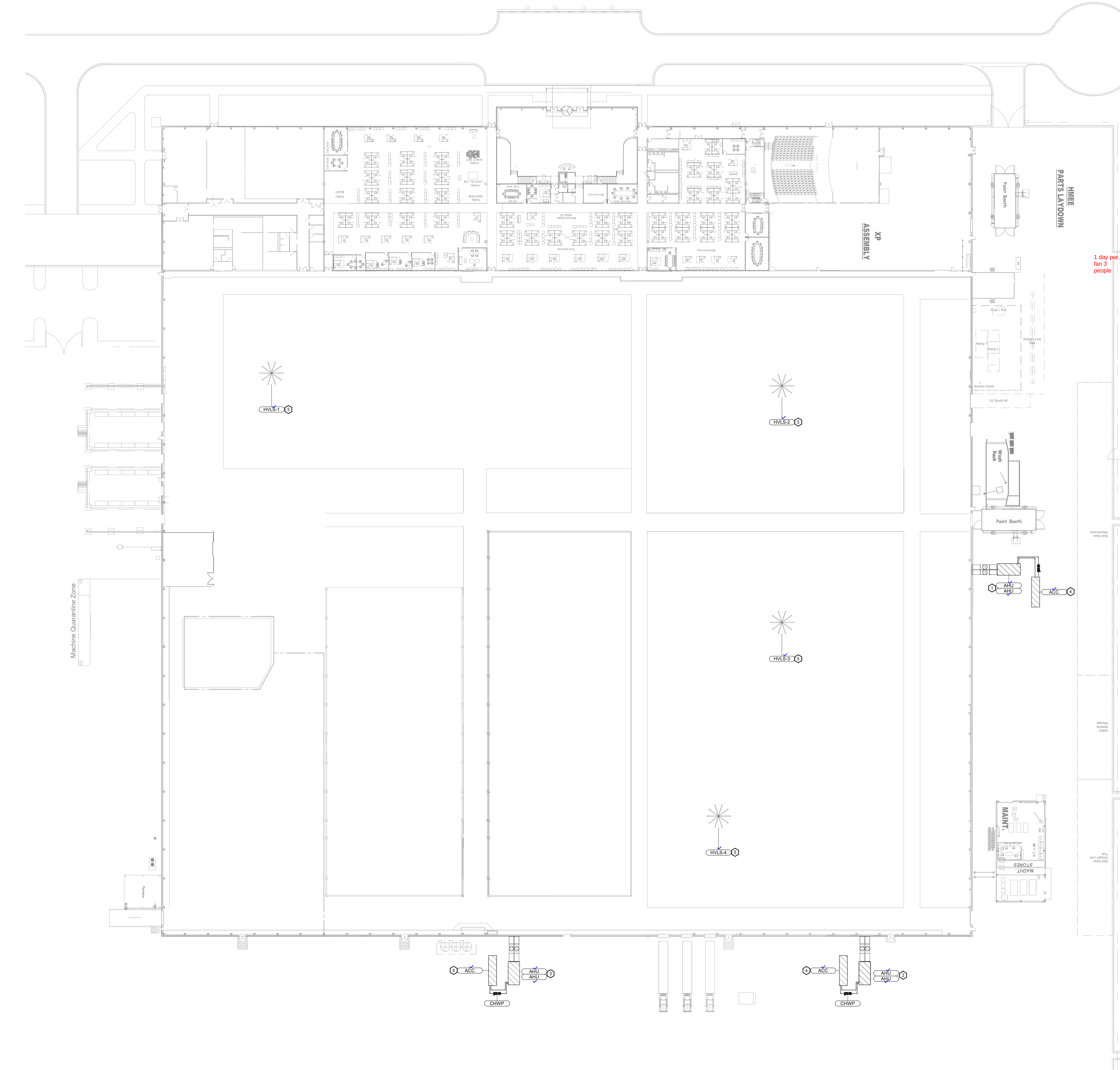
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MECHANICAL
GENERAL
SHEET

M001



CONTRACTOR TO PROVIDE TEMPORARY COOLING DURING THE DURATION OF THE PROJECT. EQUIPMENT TO BE SET UP IN LOCATIONS THAT WILL NOT INTERFERE WITH THE PERMANENT CONSTRUCTION. CONTRACTOR TO PROVIDE ALL TEMP CHILLED WATER HOSES, AND MAKEUP WATER TO ACCOMMODATE A FULLY OPERATIONAL SYSTEM.

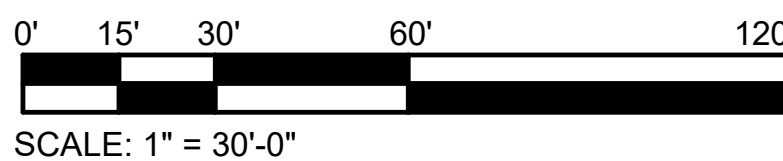
MECHANICAL KEYED NOTES:

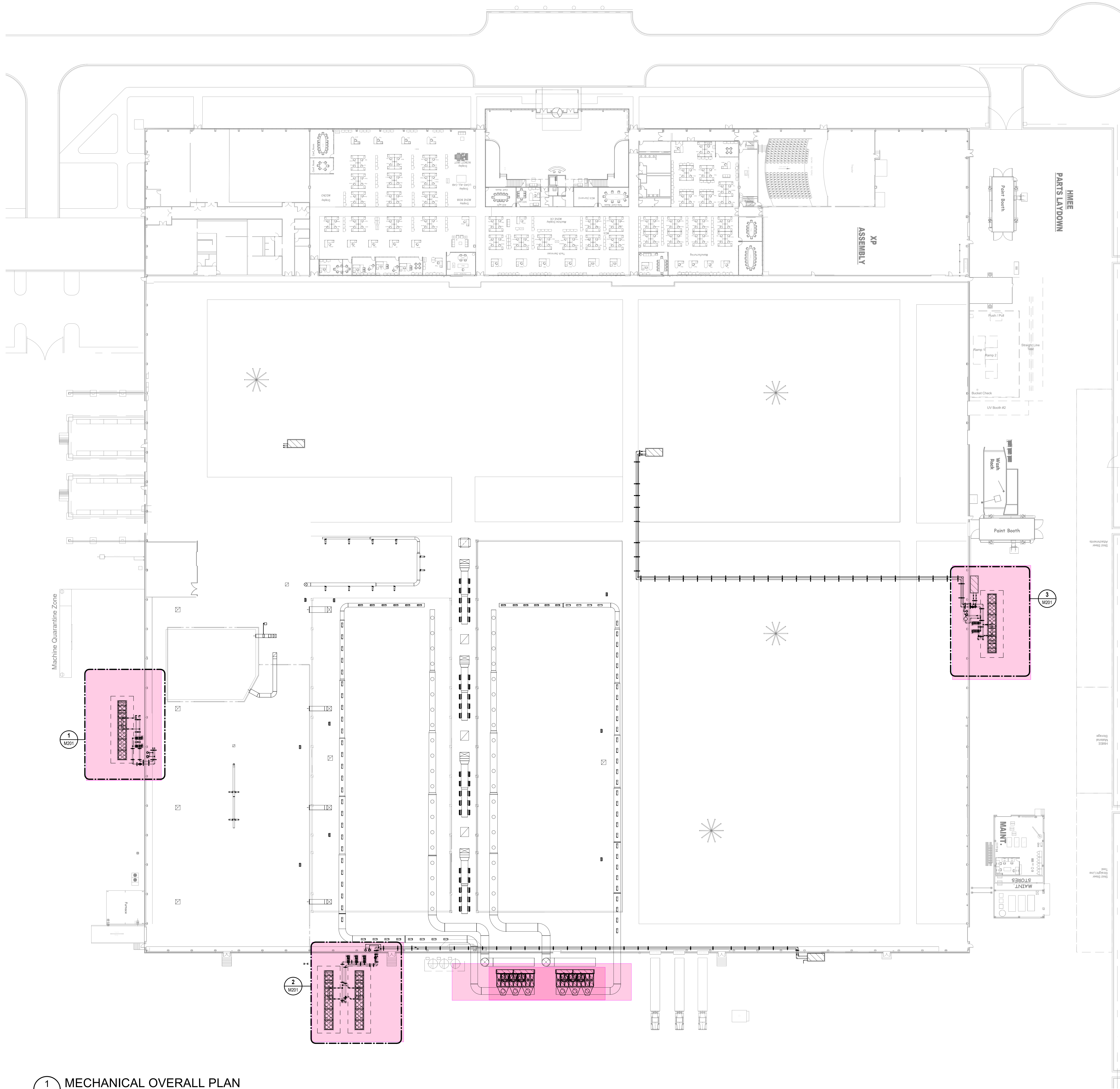
- 1 PROVIDE AND INSTALL TWO (2) VERTICALLY STACKED TEMPORARY 25,000 CFM CHILLED-WATER AIR HANDLING UNITS. ROUTE SUPPLY DUCTWORK FROM AIR HANDLERS UP ON WALL AND PENETRATE BUILDING EXTERIOR BELOW INTERIOR ROOF STRUCTURE. FUTURE PENETRATIONS TO BE USED FOR EXTERIOR AIR ROTATION UNITS.
- 2 PROVIDE AND INSTALL TWO (2) VERTICALLY STACKED TEMPORARY 25,000 CFM CHILLED-WATER AIR HANDLING UNITS. ROUTE SUPPLY DUCTWORK FROM AIR HANDLERS UP ON WALL AND PENETRATE BUILDING EXTERIOR BELOW INTERIOR ROOF STRUCTURE. FUTURE PENETRATIONS TO BE USED FOR FUTURE EXTERIOR MOUNTED AIR ROTATION UNITS.
- 3 PROVIDE AND INSTALL TWO (2) VERTICALLY STACKED TEMPORARY 25,000 CFM CHILLED-WATER AIR HANDLING UNITS. ROUTE SUPPLY DUCTWORK FROM AIR HANDLERS UP ON WALL AND PENETRATE BUILDING EXTERIOR BELOW INTERIOR ROOF STRUCTURE. FUTURE PENETRATIONS TO BE USED FOR FUTURE WELD FUME EXTRACTOR DUCTWORK.
- 4 PROVIDE AND INSTALL TEMPORARY 250 TON-MINIMUM AIR-COOLED CHILLER WITH ASSOCIATED PUMPS TO SERVE THE ADJACENT AIR HANDLING UNITS.
- 5 SUSPEND RELOCATED HVLS FAN FROM STRUCTURE ABOVE. ENSURE FAN IS CLEAR OF ALL STRUCTURE AND PLANT EQUIPMENT LOCATED BELOW. FAN WILL REMAIN FOR PERMANENT CONSTRUCTION.

CONFIRM INCLUSION IN SM SCOPE.

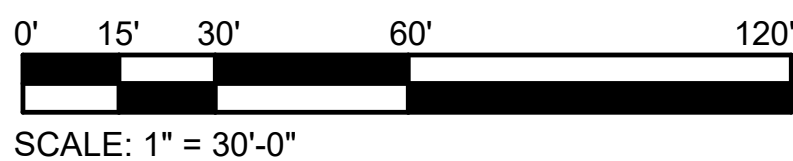
1 day per fan 3 people

1 TEMPORARY MECHANICAL OVERALL PLAN
M101 1" = 30'-0"

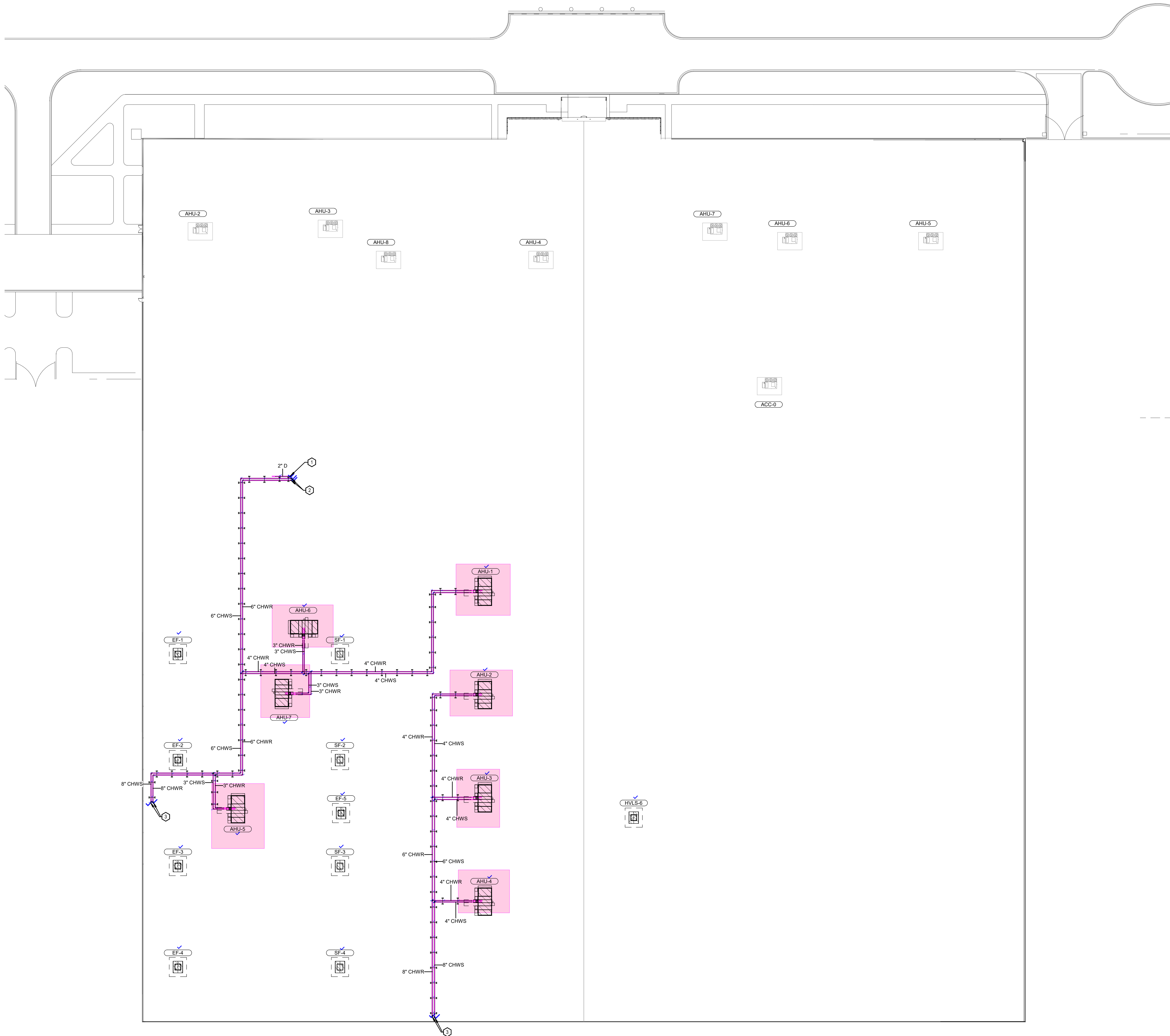




1 MECHANICAL OVERALL PLAN
M111 1" = 30'-0"

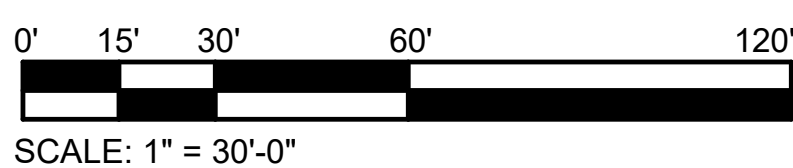


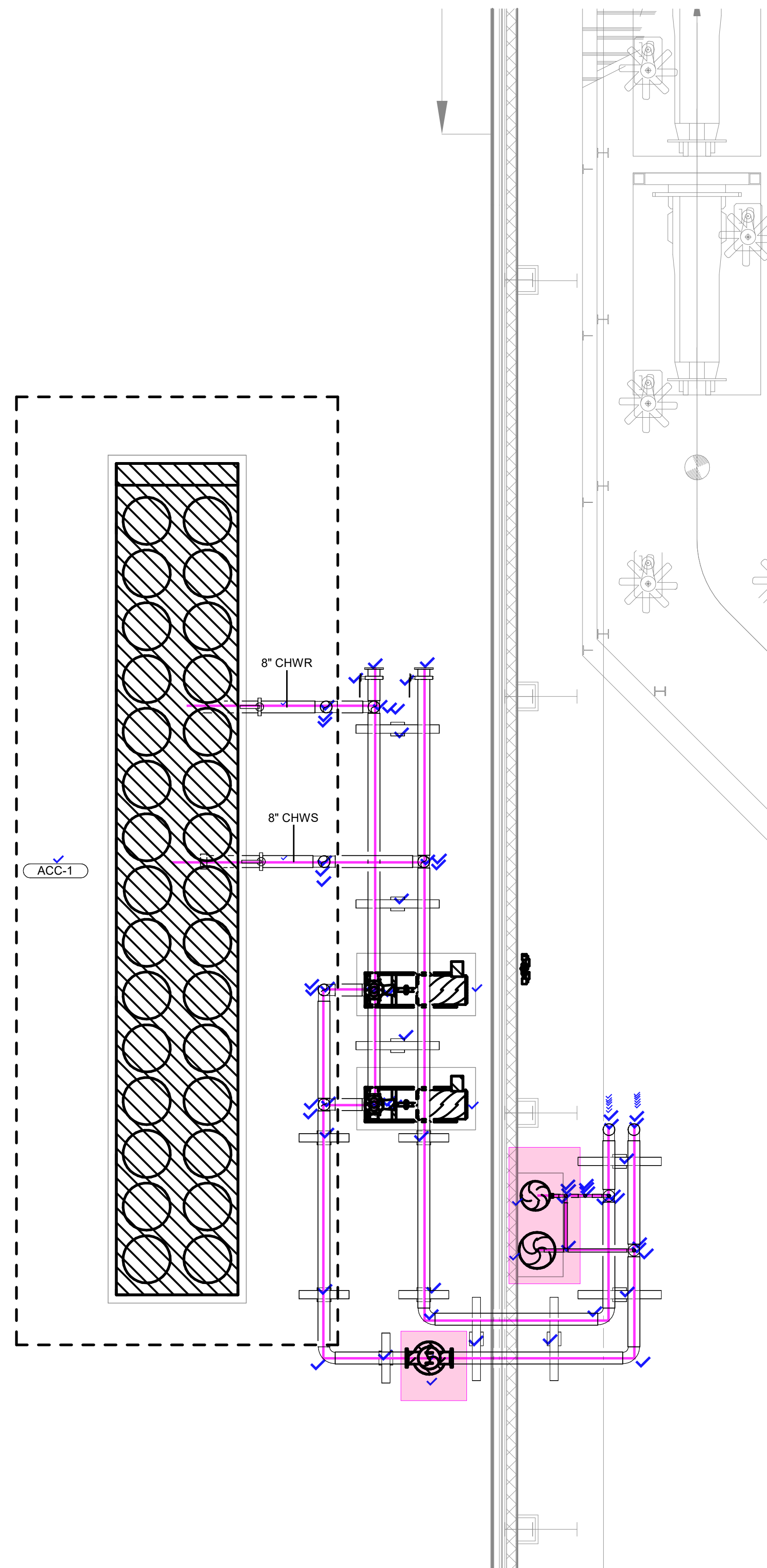
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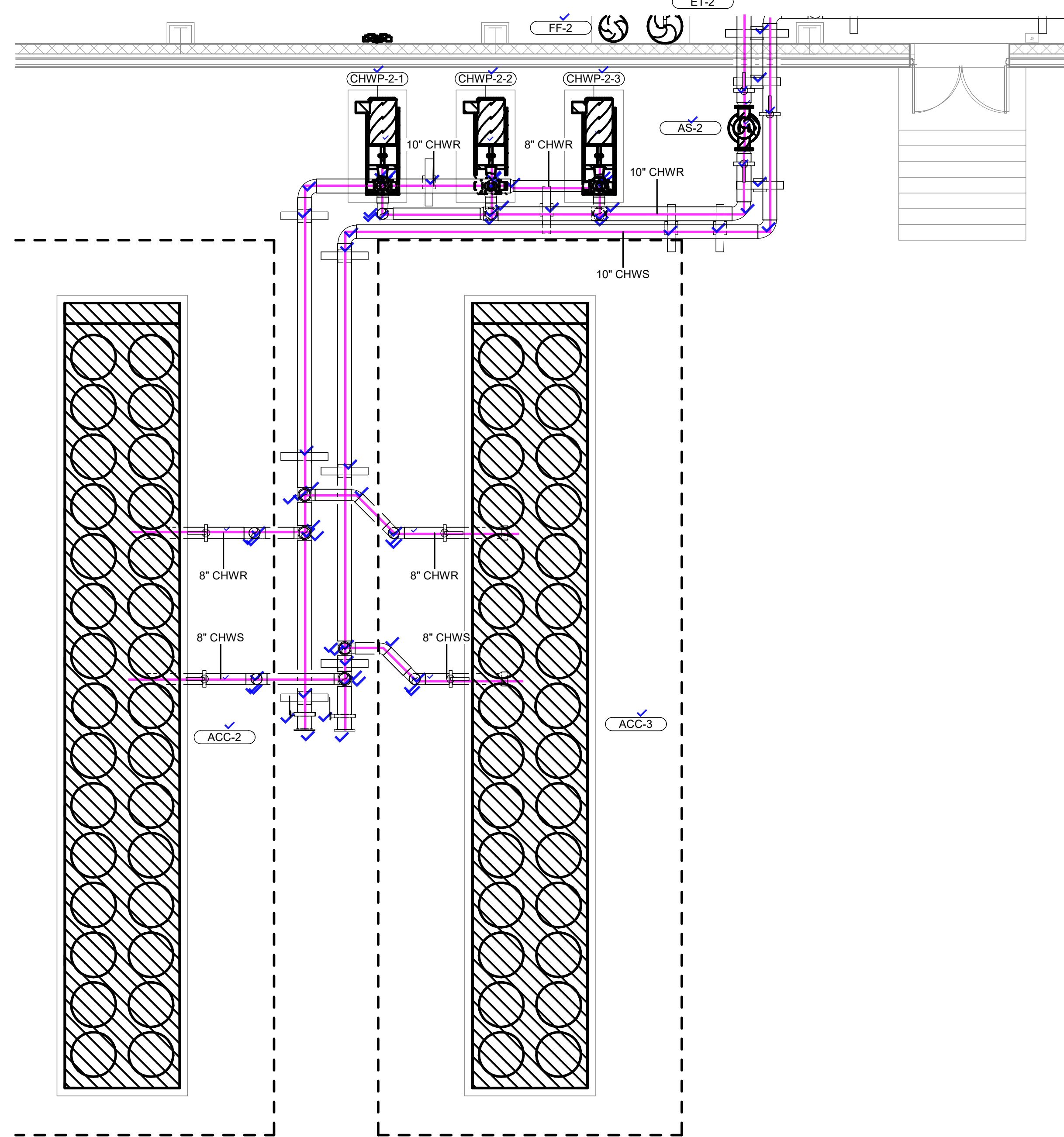
- MECHANICAL KEYED NOTES:**
- 1 INSULATED COPPER CONDENSATE UP FROM ARU BELOW. ROUTE TO NEAREST ROOF DRAIN. PROVIDE CLEANOUTS WHERE REQUIRED. TERMINATE WITH AT LEAST 1" AIR GAP.
 - 2 CHILLED WATER PIPE DOWN TO ARU BELOW. REFER TO LEVEL 1 AREA PLAN A FOR CONTINUATION.
 - 3 CHILLED WATER UP FROM LEVEL 1 BELOW. REFER TO LEVEL 1 AREA PLAN C FOR CONTINUATION.

1 MECHANICAL ROOF PLAN
M121
1" = 30'-0"

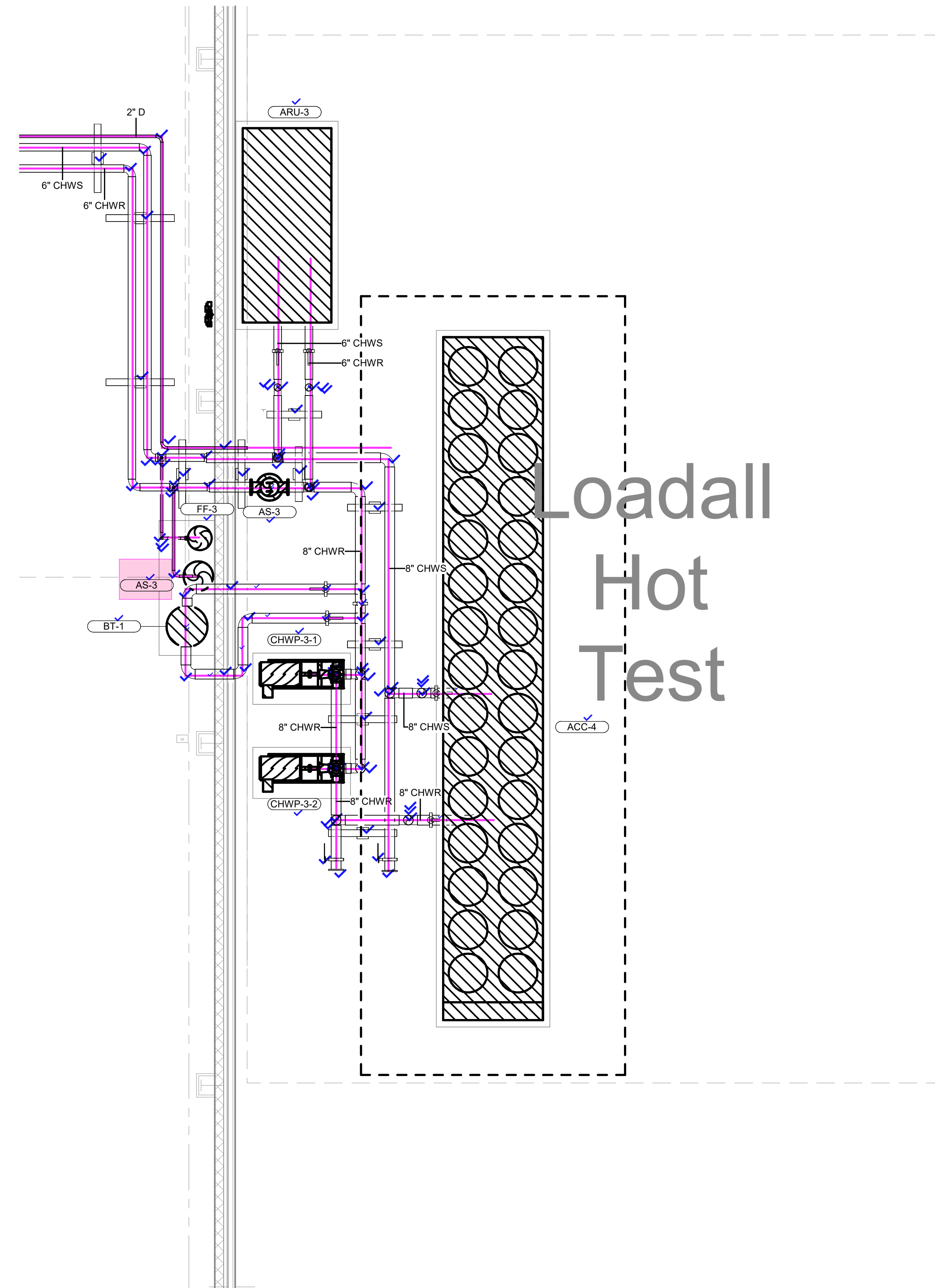




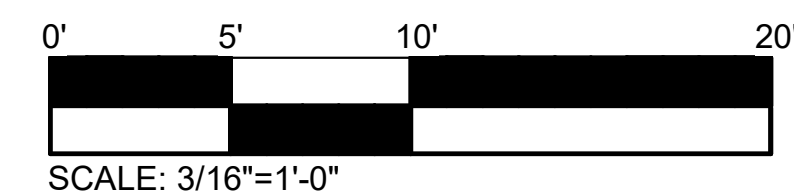
1 MECHANICAL ENLARGED PLAN - AREA C CHILLER 1
3/16" = 1'-0"



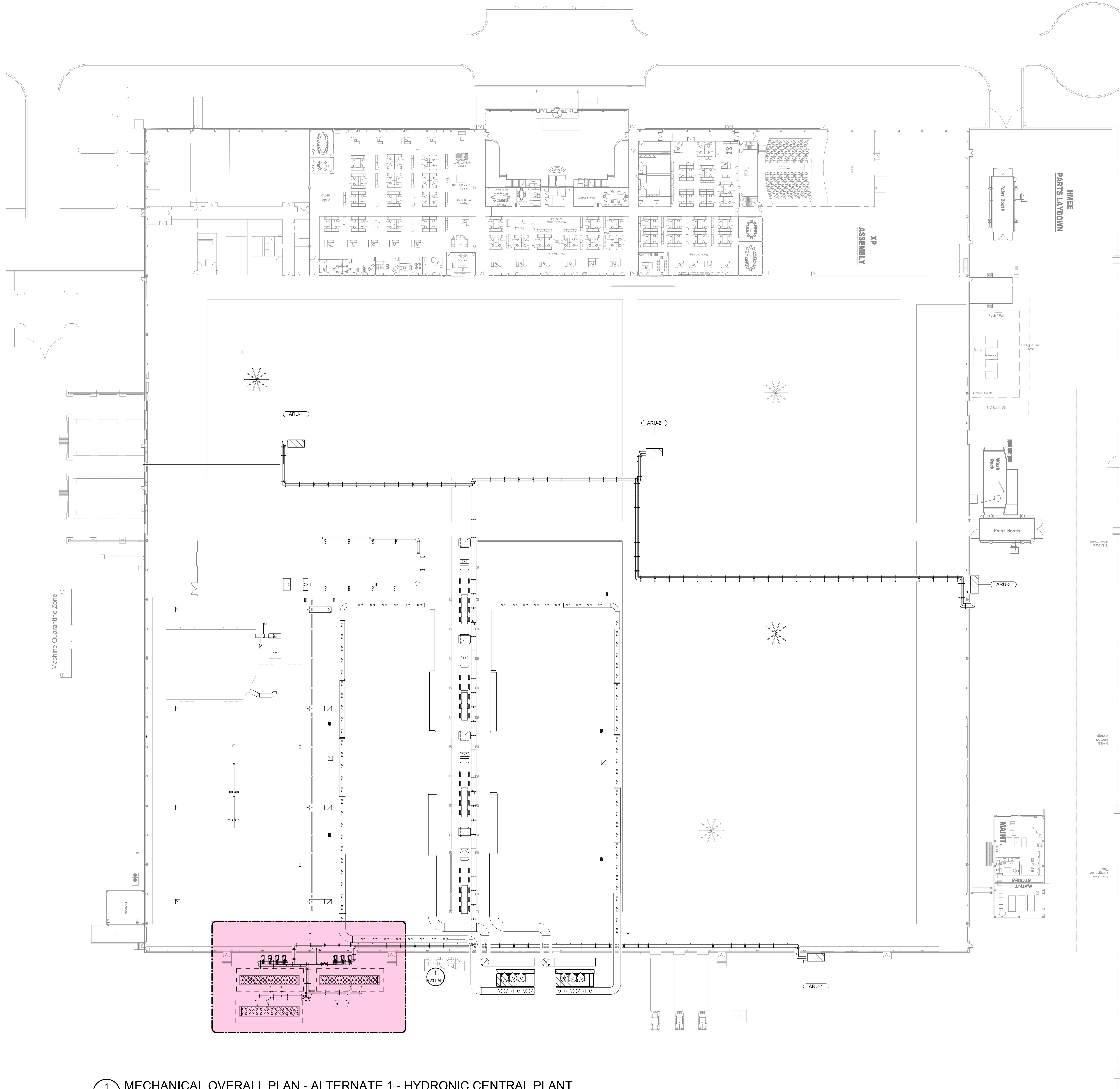
2 MECHANICAL ENLARGED PLAN - AREA C CHILLERS 2 & 3
3/16" = 1'-0"



3 MECHANICAL ENLARGED PLAN - AREA BD CHILLER 4
3/16" = 1'-0"



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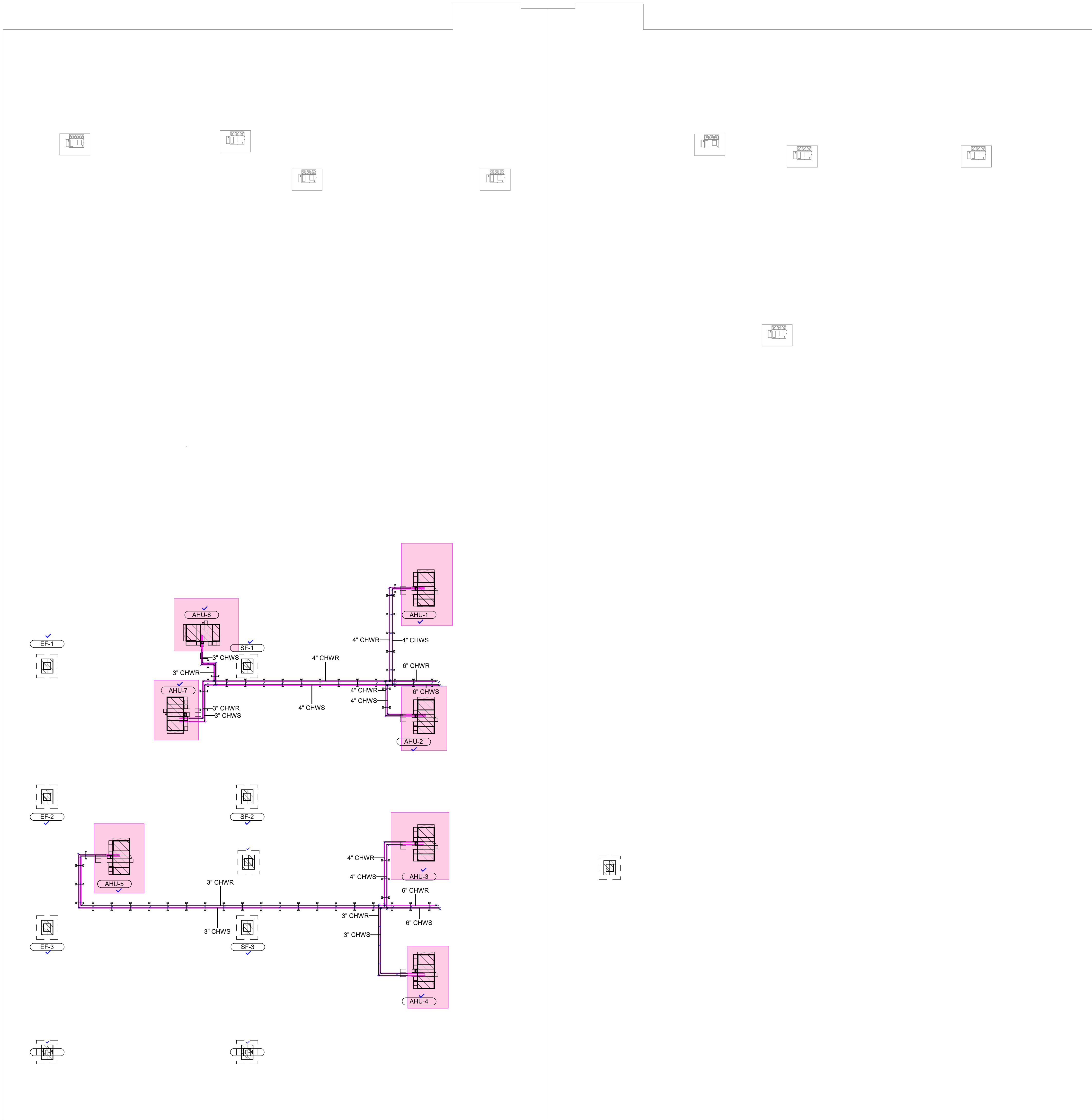


1 MECHANICAL OVERALL PLAN - ALTERNATE 1 - HYDRONIC CENTRAL PLANT
1" = 30'-0"

0' 15' 30' 60' 120'
SCALE: 1" = 30'-0"

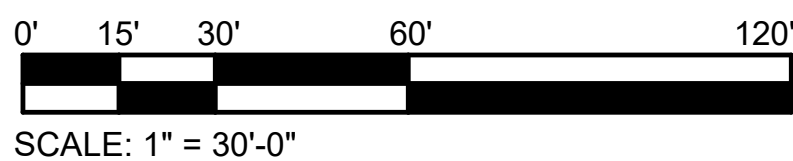
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1 MECHANICAL ROOF PLAN - ALTERNATE 1
M121-ALT 1" = 30'-0"

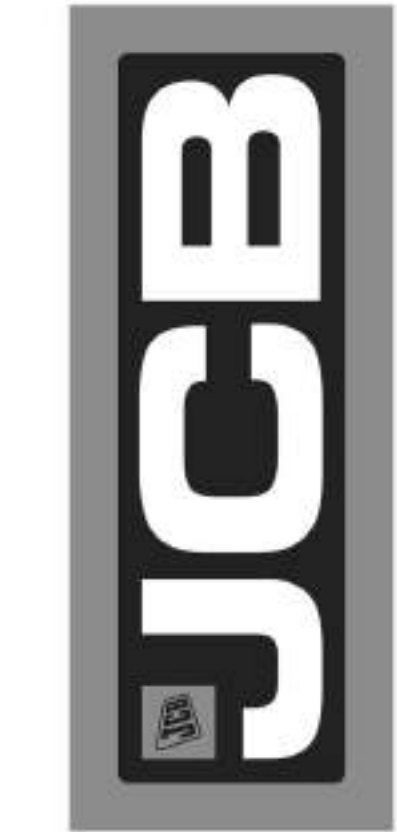


MECHANICAL GENERAL NOTES:
1. REFER TO BASE BID SHEETS FOR INFORMATION REGARDING DUCTWORK AND AIRSIDE SYSTEMS.

MECHANICAL KEYED NOTES:
① KEYED NOTE ONE.
② KEYED NOTE TWO.



SCALE: 1" = 30'-0"



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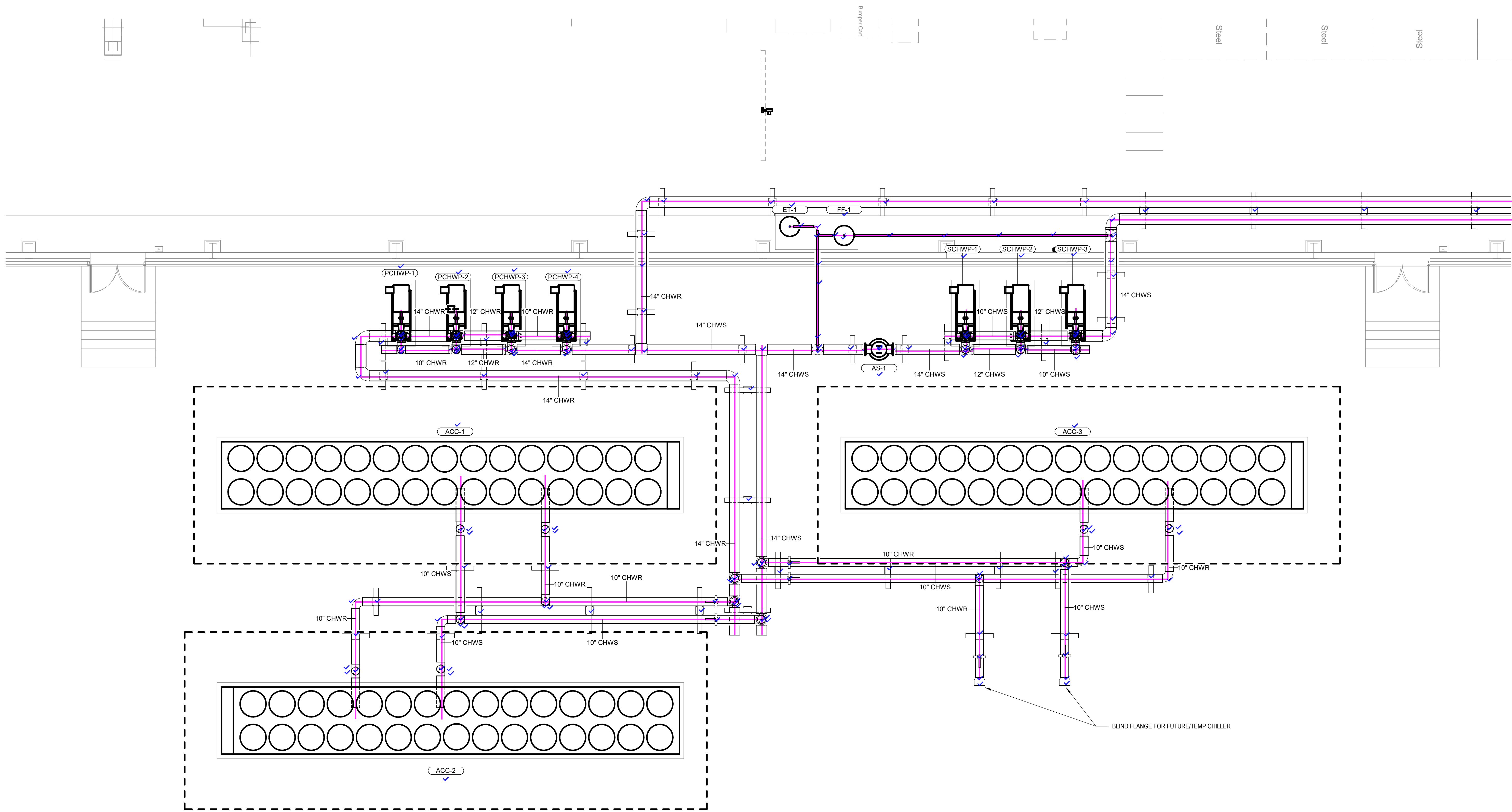
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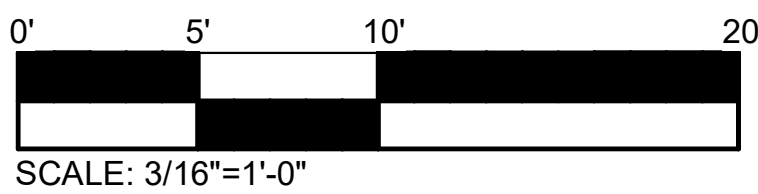
MECHANICAL
ROOF PLAN -
ALTERNATE 1

M121-ALT



MECHANICAL GENERAL NOTES:
1. REFER TO BASE BID SHEETS FOR INFORMATION REGARDING DUCTWORK AND AIRSIDE SYSTEMS

1 MECHANICAL ENLARGED PLAN - ALTERNATE 1 - HYDRONIC CENTRAL PLANT
M201-ALT 3/16" = 1'-0"



AIR-COOLED CHILLER SCHEDULE				
- BASE BID				
MARK	ACC-1	ACC-2 & ACC-3	ACC-4	
TYPE	AIR-COOLED	AIR-COOLED	AIR-COOLED	
CAPACITY (TONS) @ DEG F AMBIENT	460 @ 105	320 @ 105	445 @ 105	
REFRIGERANT	513A	513A	513A	
EER FL / (PLV @ AHR)	9.9 / 17.2	10.5 / 17.8	10.2 / 17.3	
EVAPORATOR				
CHILLED WATER-GPM	915	630	890	
WATER TEMP. (OUT °F)	42	42	42	
WATER TEMP. (IN °F)	54	54	54	
FOULING FACTOR	0.0001	0.0001	0.0001	
PRESSURE DROP MAX. FT. WATER	15.8	6.3	15.0	
CONDENSER				
DESIGN O.A. TEMP. (F)	105	105	105	
MIN. O.A. TEMP. TO CONDENSER	20	20	20	
ELECTRIC				
VOLTAGE/PHASE	480/3	480/3	480/3	
MCA - CONNECTION 1	603	734	584	
MCCP - CONNECTION 1	800	1000	700	
MCA - CONNECTION 2	594	-	557	
MCCP - CONNECTION 2	800	-	700	
GENERAL				
OPER. WEIGHT (LBS)	28,000	25,000	28,000	
MFGR. AND MODEL	DAIKIN AWY02B DAIKIN AWY02B DAIKIN AWY02B			
NOTES	1-11	1-11	1-11	

- AIR-COOLED CHILLER SCHEDULE NOTES:
- SINGLE POINT POWER CONNECTION; CONTROL XFMR AND EVAP. HTR. TO BE INTERNALLY POWERED.
 - PROVIDE WITH VFD.
 - PROVIDE WITH INTERNALLY WIRED CONTROL XFMR.
 - PROVIDE WITH SUCTION AND DISCHARGE SERVICE VALVES FOR EACH COMPRESSOR.
 - CHILLER SHALL BE MOUNTED ON NEOPRENE PADS ATOP A CONCRETE PAD. THE PAD SHALL EXTEND 4" BEYOND FOOTPRINT OF UNIT IN ALL DIRECTIONS. REFER TO STRUCTURAL.
 - PROVIDE WITH FULLY ENCLOSED LOUISVRE MAJUGARD.
 - PROVIDE WITH LOW SOUND FANS AND ATTENUATION BLANKETS FOR COMPRESSORS.
 - PROVIDE WITH MANUFACTURER'S STANDARD FIVE YEAR WARRANTY FOR COMPRESSORS.
 - PROVIDE WITH BAGNET INTERFACE.
 - PROVIDE WITH MINIMUM 60A SHORT CIRCUIT CURRENT RATING.
 - PROVIDE WITH EVAPORATOR HEAT TRACE WITH A SEPARATE 20A, 120V POINT OF CONNECTION.

AIR-COOLED CHILLER SCHEDULE	
- ALTERNATE	
MARK	ACC-1 THRU ACC-3
TYPE	AIR-COOLED
CAPACITY (TONS) @ DEG F AMBIENT	500 @ 105
REFRIGERANT	513A
EER FL / (PLV @ AHR)	9.7 / 18.0
EVAPORATOR	
CHILLED WATER-GPM	995
WATER TEMP. (OUT °F)	42
WATER TEMP. (IN °F)	54
FOULING FACTOR	0.0001
PRESSURE DROP MAX. FT. WATER	18.3
CONDENSER	
DESIGN O.A. TEMP. (F)	105
MIN. O.A. TEMP. TO CONDENSER	20
ELECTRIC	
VOLTAGE/PHASE	480/3
MCA - CONNECTION 1	698
MCCP - CONNECTION 1	1000
MCA - CONNECTION 2	695
MCCP - CONNECTION 2	1000
GENERAL	
OPER. WEIGHT (LBS)	30,000
MFGR. AND MODEL	DAIKIN AWY030B
NOTES	1-11

AIR ROTATION UNIT SCHEDULE				
TAG	ARU-1	ARU-2	ARU-3	ARU-4
AREA SERVED	ASSEMBLY	ASSEMBLY	ASSEMBLY	ASSEMBLY
UNIT HEIGHT (FT)	30	30	30	30
EXT. STATIC PRESS (IN. W.C.)	0.32	0.32	0.32	0.32
FAN DRIVE	ECM	ECM	ECM	ECM
MAX FAN BHP (EACH FAN)	10.71	12.71	12.56	12.56
FAN MOTOR (HP/QUANTITY)	2 / 15	2 / 15	2 / 15	2 / 15
FAN TYPE	PLENUM	PLENUM	PLENUM	PLENUM
FAN WALL ARRANGEMENT	2X1	2X1	2X1	2X1
MINIMUM OA CFM	-	-	9,170	9,170
CFM	58,450	58,450	58,450	58,450
COOLING SECTION				
COOLING COIL TYPE	CHW	CHW	CHW	CHW
ENT AIR (DBWB)	80 / 70	80 / 70	83 / 72	83 / 72
LEAVING AIR (DBWB)	58 / 57	58 / 57	58 / 57	58 / 57
MAX FACE VEL. (FPM)	500	500	500	500
MAX AIR S.P.D. (IN. WATER)	0.375	0.316	0.366	0.366
GPM F12 TEMP. RISE	416	413	476.5	476.5
MAX WATER PD (FT)	7.148	8.531	7.418	7.418
ROWS/FMS (PER IN)	4 / 9	3 / 12	4 / 11	4 / 11
TOTAL OUTPUT (MIN. MBH)	2954	2484	2870	2870
SENS. OUTPUT (MIN. MBH)	1408	1999	1572	1572
HEATING SECTION				
HEATING COIL TYPE	NAT GAS	-	NAT GAS	NAT GAS
MAX GAS OPERATING PRESSURE	5 PSI	-	5 PSI	5 PSI
ENT AIR (DB)	65	-	65	65
LEAVING AIR (DB)	85	-	85	85
TOTAL INPUT (MIN. MBH)	1800	-	1800	1800
TOTAL OUTPUT (MIN. MBH)	1440	-	1440	1440
INLET SECTION				
FILTER TYPE	2" PLEATED	2" PLEATED	2" PLEATED	2" PLEATED
FILTER MERV	MERV 8	MERV 8	MERV 8	MERV 8
VOLTAGE/PH	460/3	460/3	460/3	460/3
MCA/MCCP	52 / 60	52 / 60	52 / 60	52 / 60
MANUFACTURER	JOHNSON AIR ROTATION	JOHNSON AIR ROTATION	JOHNSON AIR ROTATION	JOHNSON AIR ROTATION
NOTES	1,2,3,4,5,6	1,2,3,4,5,6	1,2,3,4,5,6	1,2,3,4,5,6

- AIR ROTATION UNIT SCHEDULE NOTES:
- PROVIDE WITH EQUIPMENT CONSTRUCTION FOR OUTDOOR USE.
 - PROVIDE WITH ECM SUPPLY FAN ARRAY.
 - PROVIDE OA / RA OPENINGS WITH INTEGRAL MOTORIZED DAMPERS.
 - PROVIDE WITH MANUFACTURER CONTROLLER WITH BAGNET COMPATABILITY.
 - PROVIDE WITH DUCTED SUPPLY AND RETURN CONNECTIONS.
 - PROVIDE UNIT WITH 5-1/2" TALL HOUSEKEEPING PAD THAT EXTENDS 4" BEYOND THE UNIT ON ALL 4 SIDES.

WELD FUME EXTRACTION SCHEDULE

MARK	TYPE	MANUFACTURER & MODEL	SERVICE	CFM	TOTAL S.P. (W.G)	CA CONNECTION	FAN TYPE	MERV RATING	CFM	E.S.P. (W.G)	MAX. BHP	H.P.	FAN QUANTITY	VPH	DRIVE	NOTES
WE-1	CARTRIDGE FILTER	CAMFL GS72	WELD FUME EXTRACTION	75000	13"	1-1/2"	CENTRIFUGAL	15	75000	7.5"	77	100	4	480/3	VFD	1,2,3,4,5,6,7
WE-2	CARTRIDGE FILTER	CAMFL GS72	WELD FUME EXTRACTION	75000	13"	1-1/2"	CENTRIFUGAL	15	75000	7.5"	77	100	4	480/3	VFD	1,2,3,4,5,6,7

- WELD FUME EXTRACTION SCHEDULE NOTES:
- PROVIDE WITH SINGLE POINT POWER CONNECTION TO MANUFACTURER CONTROL PANEL LOCATED INDOORS.
 - PROVIDE FILTER AND FAN HOUSING WITH RAISED HOUSEKEEPING PAD BELOW SUPPORTS.
 - PROVIDE WITH MFG PROVIDED FILTER ACCESS PLATFORM AND LADDER WITH GAGE.
 - COMPRESSED AIR REQUIREMENTS: 60 SCFM AT 90PSI.
 - FANS SHALL BE MOUNTED ON TOP OF EQUIPMENT AT 45° OFFSET AND DISCHARGE TO THE OPPOSITE SIDE OF THE EQUIPMENT INLET CONNECTIONS.
 - EACH FAN SHALL BE PROVIDED WITH INDIVIDUAL VFD PROVIDED BY EQUIPMENT MANUFACTURER AND LOCATED IN THE CONTROL CABINET.
 - PROVIDE WITH BAGNET CONTROLLER. EQUIPMENT TO BE CONTROLLED BY MFG PROVIDED CONTROLLER. DDC TO MONITOR ALL POINTS PROVIDED BY MFG CONTROLLER INCLUDING BUT NOT LIMITED TO: DIRTY FILTER STATUS, VFD START/STOP (PER FAN), VFD FAULT 1 (PER FAN), VFD SPEED (PER FAN)

HUMIDIFIER SCHEDULE	
MARK	GHU-1A & B
CFM	8,500
DESIGN CONDITION	
SPACE (DB/HR)	70 / 50
ENTERING (DB/HR)	85 / 33
TYPE	NAT GAS
STEAM (LBS/HR)	46
MAX ENERGY INPUT (BTU/HR)	62,000
ELECTRICAL (V/PH/MCA/MCCP)	120/60/215
REFERENCE	CONDAR GS 50-CS
NOTES	1-4

- HUMIDIFIER SCHEDULE NOTES:
- PROVIDE MANUFACTURER WALL MOUNT BRACKET / FRAME.
 - PROVIDE DUCT MOUNTED HUMIDISTAT, HIGH LIMIT CUTOFF, AND WALL MOUNTED HUMIDISTAT WITH 0-15V CONTROL.
 - PROVIDE WITH DISPERSION PIPING ROUTED TO DUCT MOUNTED HUMIDIFIER.
 - UNIT MOUNTED IN HORIZONTAL DUCT WITH UPFLOW SUPPLY AIR.

BUFFER TANK SCHEDULE	
TAG	BT-1
SYSTEM	CHILLED WATER
DIAMETER (IN)	30"
APPROX. VOL. GAL.	350
INLET / OUTLET (SIZE - IN.) - FLANGED	4"
MANUFACTURER	TACO
MODEL	BTH0350F-129NN
WEIGHT LBS (EMPTY/FILLED)	910 / 3626
NOTES:	1,2,3,4,5,6,7

- BUFFER TANK SCHEDULE NOTES:
- FACTORY INSULATE TANK WITH 2" INSULATION.
 - MOUNT ON 3.5' CONCRETE HOUSEKEEPING PAD.
 - CONSTRUCTED/TESTED TO ASME SECTION VII, DIVISION I.
 - TANK TO BE INTERNALLY BAFFLED WITH HIGH FLANGED INLETS AND OUTLETS (150 #).
 - 1" DRAIN ON UNDERSIDE OF TANK.
 - 125 PSI WORKING PRESSURE AT 200 DEG F. ASME RATED.

PUMP SCHEDULE - BASE BID				
MARK	SERVICE	TYPE	GPM	HEAD (FT)
CHWP-1-1	CHW	END SUCTION	915	105
CHWP-1-2	CHW	END SUCTION	915	105
CHWP-2-1	CHW	END SUCTION	630	110
CHWP-2-2	CHW	END SUCTION	630	110
CHWP-2-3	CHW	END SUCTION	630	110
CHWP-3-1	CHW	END SUCTION	890	105
CHWP-3-2	CHW	END SUCTION	890	105

- PUMP SCHEDULE NOTES:
- PUMP TO BE MOUNTED ON INERTIA BASE ON TOP OF CONCRETE HOUSEKEEPING PAD PER DETAILS.
 - MECHANICAL CONTRACTOR TO PROVIDE AND INSTALL VFD. ELECTRICAL CONTRACTOR TO CONNECT FROM PANEL TO VFD AND TO MOTOR.

PUMP SCHEDULE - ALTERNATE				
MARK	SERVICE	TYPE	GPM	HEAD (FT)
POHWP-1	PRIMARY CHW	END SUCTION	990	45
POHWP-2	PRIMARY CHW	END SUCTION	990	45
POHWP-3	PRIMARY CHW	END SUCTION	990	45
POHWP-4	PRIMARY CHW	END SUCTION	990	45
SCHWP-1	SECONDARY CHW	END SUCTION	1530	100
SCHWP-2	SECONDARY CHW	END SUCTION	1530	100
SCHWP-3	SECONDARY CHW	END SUCTION	1530	100

- PUMP SCHEDULE NOTES:
- PUMP TO BE MOUNTED ON INERTIA BASE ON TOP OF CONCRETE HOUSEKEEPING PAD PER DETAILS.
 - MECHANICAL CONTRACTOR TO PROVIDE AND INSTALL VFD. ELECTRICAL CONTRACTOR TO CONNECT FROM PANEL TO VFD AND TO MOTOR.

DUCT HEATER SCHEDULE

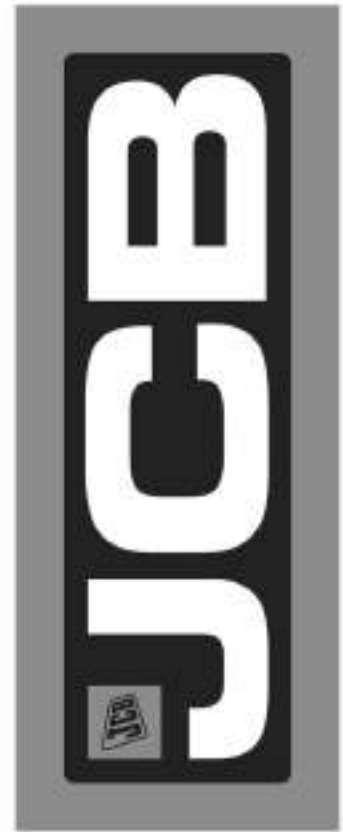
MARK	EDH-1
HEATER TYPE	ELECTRIC
CFM	8,500
ENTERING AIR (°F DB)	63
LEAVING AIR (°F DB)	85
MIN. MBU	202
MIN. KW	60
ELECTRICAL (VOLT/PH)	480/3
CONTROL CIRCUIT VOLTAGE	24V
DUCT DIMENSIONS (INSIDE)	
WIDTH (IN.)	36
HEIGHT (IN.)	26
AIRFLOW INTERLOCK METHOD	DIFF. PRESSURE
CONTROL OPTION	SCR
MANUFACTURER	REDD-I
NOTES	1

- DUCT HEATER SCHEDULE NOTES:
- FURNISH WITH DOOR INTERLOCK DISCONNECT, SCR, DIFF. PRESSURE SWITCH

GAS UNIT HEATER SCHEDULE

MARK	CFM	ELECT. (WFLA/MCCP)	INPUT (BTU/H)	OUTPUT (BTU/H)	MOTOR HP	GAS CONNX.	COMBUSTION AIR DIAM.	VENT DIAM.	REFERENCE	NOTES
GUH-1	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-2	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-3	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-4	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-5	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-6	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-7	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-8	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4
GUH-9	5123	115/1120	400,000	332,000	0.5	3/4"	6"	6"	REZNOR UD2-400	1,2,3,4

- ELECTRIC UNIT HEATER SCHEDULE NOTES:
- PROVIDE MANUFACTURER'S STD. WALL MOUNTED THERMOSTAT, SET AT 50 DEGREES.
 - PROVIDE MANUFACTURER'S STD. MOTOR STARTER.
 - PROVIDE WITH MANUFACTURER'S CONCENTRIC COMBINATION INTAKE/FLUE VENT KIT.
 - EQUIPMENT SHALL BE INSTALLED IN COMPLIANCE WITH 2021 IMC 600.1



PRELIMINARY
REVIEW SET
NOT FOR REGULATORY
APPROVAL, PERMITTING, OR
CONSTRUCTION
WADE A. CLEARY
P.E. REG. NO. 027730
DEC 2, 2025

JCB SAVANNAH PLANT - AIR QUALITY UPGRADES

2000 BAMFORD BLVD., POOLER, GA 31322

Project Number 250829(cz)
Date 12/02/2025
Drawn By CZT
Checked By Checker

Revisions
No. Description Date

PLUMBING
SYMBOLS &
ABBREVIATIONS

P000

PLUMBING SYMBOLS AND ABBREVIATIONS

NOTE: SELDOM ARE ALL SYMBOLS AND ABBREVIATIONS USED IN THE DRAWINGS. REFERENCE ONLY THOSE THAT ARE APPLICABLE.

SYMBOL	DESCRIPTION
	GAS COCK OR PLUG VALVE
	PRESSURE RELIEF VALVE
	TEMPERATURE AND PRESSURE RELIEF VALVE
	VACUUM BREAKER
	FLOW SWITCH
	FLOOR DRAIN, (TYPE)
	FLOOR SINK, (TYPE)
	PRESSURE SWITCH
	GATE VALVE
	TAMPER SWITCH
	O.S.&Y VALVE
	SOLENOID VALVE
	POST INDICATOR VALVE
	SWING CHECK VALVE
	NON-SLAM CHECK VALVE
	BALL VALVE
	BUTTERFLY VALVE
	PIPE RISE (R) OR DROP (D)
	FLOW - IN DIRECTION OF ARROW
	CAP ON END OF PIPE
	CONCENTRIC REDUCER

SYMBOL	DESCRIPTION
	YARD CLEANOUT OR CLEANOUT TO GRADE
	VALVE IN RISER (TYPE AS SPEC'D OR NOTED)
	PLUMBING FIXTURES
	NOTES (NEW CONSTRUCTION)
	DEMOLITION NOTES
	EQUIPMENT IDENTIFICATION
	PLUMBING FIXTURE AND EQUIPMENT MARK
	PLUMBING RISER
	GAS METER
	UNION (FLANGED)
	CLEANOUT PLUG
	FLOOR CLEANOUT
	CONNECT TO EXISTING (PROVIDE AND INSTALL ALL NECESSARY TRANSITION FITTINGS)
	DETAIL REFERENCE NUMBER ON SHEET
	SHEET NUMBER
	PRIMARY ROOF DRAIN (OUTLET SIZE)
	SECONDARY (EMERGENCY) OVERFLOW DRAIN/ ROOF DRAIN (OUTLET SIZE)

PLUMBING GENERAL NOTES (APPLY TO ALL SHEETS):

1. ALL MATERIALS AND CONSTRUCTION PROCEDURES PERTAINING TO THE WORK DESCRIBED IN THESE DRAWINGS SHALL CONFORM TO THE REQUIREMENTS OF ALL AUTHORITIES HAVING JURISDICTION.
2. CONTRACTORS SHALL OBTAIN AND MAKE PROVISION FOR ALL PERMITS, INSPECTIONS, AND TESTS REQUIRED BY AUTHORITIES HAVING JURISDICTION.
3. CONTRACTORS SHALL BE SOLELY RESPONSIBLE FOR VERIFYING ACTUAL CONDITIONS AT THE SITE AND NOTING ALL DISCREPANCIES TO THE OWNER PRIOR TO WORK COMMENCEMENT. THEREAFTER, THE CONTRACTOR ACCEPTS FULL RESPONSIBILITY FOR ALL EXISTING CONDITIONS AND SHALL BE SOLELY RESPONSIBLE FOR MAKING ALL SUITABLE ADJUSTMENTS NECESSARY TO ACCOMMODATE NEW WORK AT NO ADDITIONAL COST TO THE OWNER. ANY SUCH ADJUSTMENTS SHALL BE COORDINATED WITH THE OWNER AND ARCHITECT.
4. CONTRACTORS SHALL INCORPORATE ALL DISCREPANCIES AND ADJUSTMENTS INTO THE CONSTRUCTION DOCUMENTS.
5. CONTRACTORS SHALL COORDINATE ALL WORK WITH OTHER TRADES AND INCLUDE ALL NECESSARY MODIFICATIONS TO ACCOMMODATE THEIR WORK.
6. CONTRACTORS SHALL COORDINATE ALL WORK WITH THE OWNER.
7. ALL MATERIALS AND WORKMANSHIP SHALL BE GUARANTEED FOR A MINIMUM OF ONE YEAR FROM THE DATE OF INSTALLATION.
8. CONTRACTORS SHALL BE SOLELY RESPONSIBLE FOR THE SAFETY OF THEIR EMPLOYEES AND SUBCONTRACTORS AND ALL OTHER PERSONS IN THE AREAS OF CONSTRUCTION. CONTRACTORS SHALL ALSO BE RESPONSIBLE FOR THE SAFETY OF ALL PROPERTY BEING ERECTED.
9. PLUMBING SERVICES THAT INTERFERE WITH ANY NEW ARCHITECTURAL WORK SHALL BE RELOCATED AS NECESSARY.

SYMBOL	DESCRIPTION
	COLD WATER PIPING
	HOT WATER PIPING
	HOT WATER RETURN PIPING
	120° HOT WATER PIPING
	140° HOT WATER PIPING
	120° HOT WATER RETURN PIPING
	140° HOT WATER RETURN PIPING
	SOFT WATER PIPING
	DEIONIZED WATER PIPING
	TEMPERED WATER PIPING
	FILTERED WATER PIPING
	COMPRESSED AIR PIPING
	MEDICAL AIR PIPING
	MEDICAL VACUUM PIPING
	MEDICAL OXYGEN PIPING
	MED NITROUS OXIDE PIPING
	MED NITROGEN PIPING
	WASTE ANESTHETIC GAS DISPOSAL PIPING
	COMBUSTION AIR EXHAUST
	COMBUSTION AIR INTAKE

SYMBOL	DESCRIPTION
	WASTE PIPING (ACID RESISTANT)
	VENT PIPING (ACID RESISTANT)
	FIRE PROTECTION PIPING
	REMOTE FDC PIPING
	AUTOMATIC SPRINKLER
	GREASE WASTE PIPING
	GREASE VENT PIPING
	TRAP-PRIMER PIPING (1/2" COPPER)
	PUMPED DRAIN PIPING
	SANITARY SEWER PIPING (WASTE)
	VENT PIPING
	DRAIN PIPING
	DRAIN VENT PIPING
	GAS REGULATOR VENT PIPING
	NATURAL GAS PIPING
	NATURAL GAS PIPING (MED PRESS)
	ROOF DRAIN PIPING (PRIMARY SYSTEM)
	ROOF DRAIN PIPING (SECONDARY 'EMERGENCY OVERFLOW DRAIN' SYSTEM)
	SITE STORM DRAIN PIPING

NOTE: EXISTING PIPING, FITTINGS AND EQUIPMENT WILL BE INDICATED WITH A LIGHTER LINE WEIGHT THAN NEW WORK.

SYMBOL	DESCRIPTION
ABV	ABOVE
AC	ABOVE CEILING
AD	ACCESS DOOR
ADA	AMERICANS WITH DISABILITIES ACT
ADJUST	ADJUSTABLE
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AP	ACCESS PANEL
ASSY	ASSEMBLY
AV	AIR VENT
AVTR	ACID VENT THRU ROOF
BF	BELOW FLOOR
BV	BALL VALVE
CI	CAST IRON
CLG	CEILING
CO	CLEANOUT
CONG	CONCRETE
COND	CONDENSATE
CONN	CONNECTION
CONT	CONTINUATION
C.W.B.F.	COLD WATER PIPING BELOW FLOOR
C.W.O.H.	COLD WATER PIPING OVERHEAD
DEMO	DEMOLISH
DIAG	DIAGRAM
D	DRAIN
D.I.	DUCTILE IRON
DN	DOWN
DV	DRAIN VENT
DWG, DRWG	DRAWING
EW	ELECTRIC WATER HEATER
ELECT	ELECTRICAL
ELEV	ELEVATION
EO	EMERGENCY OVERFLOW DRAIN
EXIST	EXISTING
EXT	EXTENTION
FO	FLOOR CLEANOUT
FD	FLOOR DRAIN
F.F.	FINISHED FLOOR
FN	FINISHED
FLR	FLOOR
FS	FLOOR SINK
FT	FEET
GAL(S)	GALLON(S)
GALV	GALVANIZED
GPM	GALLONS PER MINUTE
GTRV	GREASE VENT THRU ROOF
GWH	GAS WATER HEATER
GV	GREASE VENT

SYMBOL	DESCRIPTION
HB	HOSE BIBB
HC	HANDICAPPED
HD	HUB DRAIN
HTR	HEATER
H.W.B.F.	HOT WATER PIPING BELOW FLOOR
H.W.O.H.	HOT WATER PIPING OVERHEAD
INV	INVERT
IN	INCHES
MAX	MAXIMUM
MECH	MECHANICAL
MIN	MINIMUM
MTD	MOUNTED
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN
NTS	NOT TO SCALE
OFD	OVERFLOW ROOF DRAIN
OH	OVERHEAD
PA	Pipe ANCHOR
PD	PRESSURE DROP
PLBG	PLUMBING
PRESS	PRESSURE
PSI	POUNDS PER SQUARE INCH, GAUGE
PVC	POLYVINYL CHLORIDE
REC	RECEIVED
REQ	REQUIRED
RD	ROOF DRAIN
RPBP	REDUCED PRESSURE BACKFLOW PREVENTER
SA	SHOCK ARRESTER
SHT	SHEET
SPEC(S)	SPECIFICATION(S)
S	SANITARY WASTE
SS	SANITARY SEWER
STAT	STATIC
TEMP	TEMPERATURE
TMV	THERMOSTATIC MIXING VALVE
TP	TRAP PRIMER OR T.P. SUPPLY
TYP	TYPICAL
UG	UNDERGROUND
V	VENT
VCP	VITRIFIED CLAY PIPE
VIR	VALVE IN RISER
VTR	VENT THROUGH ROOF
WA	WATER HAMMER ARRESTER
WH	WALL HYDRANT
WI	WITH
W/O	WITHOUT
WCO	WALL CLEANOUT
YCO	YARD CLEANOUT

1 PLUMBING FLOOR PLAN - OVERALL
P111 1" = 30'-0"

